

Balázs Székér

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Education

Master of Science in Statistics

2024–2027

ETH Zürich (GPA: 5.55/6.0)

Zürich, Switzerland

- **Semester Paper and Master Thesis** on Continual Learning in Multi-Turn Agentic Tool-Use Environments.
- **Relevant Coursework:** Advanced Machine Learning, Probabilistic Artificial Intelligence, Big Data, Foundations of Reinforcement Learning, Large Language Models.

Bachelor of Science in Physics

2019–2022

ETH Zürich (GPA: 5.25/6.0)

Zürich, Switzerland

- **Relevant Coursework:** Linear Algebra, Analysis

Experience

AI Engineering Intern (LLMs & Retrieval Systems)

Oct 2025 – Mar 2026

United Nations Office for Disaster and Risk Reduction

Bangkok, Thailand

- Project Manager leading the design and development of a RAG-based LLM system for large-scale knowledge access.
- Designed a Python ingestion and preprocessing pipeline (JSON API), including document chunking strategies.
- Experimented with retrieval configurations to improve relevance and factual grounding.
- Implemented embedding and vector search (OpenAI API, PostgreSQL, Azure AI Search).
- Built a chatbot prototype for internal evaluation and deployment.

Data Scientist (Spatiotemporal Modelling)

Mar 2025 – Jul 2025

Sika Technology AG, Research and Development

Zürich, Switzerland

- Designed and implemented an application to forecast global roof-color evolution based on satellite imagery using spatiotemporal regression.
- Developed a Python pipeline leveraging the Google Earth API to retrieve high-resolution satellite images.
- Adapted a physics-based correction model to remove atmospheric and surface-impurity artifacts from raw satellite data.
- Engineered a time-series regression framework to capture and predict roof-color changes over time.
- Employed Gaussian Process Regression to spatially model and optimize the hyperparameters governing the temporal evolution model.

Research Intern (Computer Vision)

Oct 2022 – Dec 2022

EMPA - Swiss Federal Laboratories for Material Science and Technology

Dübendorf, Switzerland

- Developed a Python pipeline for vehicle detection and tracking in drone footage (optical flow), producing velocity/acceleration profiles for downstream analysis.

Teaching Assistant (Machine Learning)

Feb 2025 – Present

ETH Zürich

Zürich, Switzerland

- Probabilistic Artificial Intelligence (A. Krause, Master's-level course): Responsible for the creation and revision of coding assignments.
- Introduction to Machine Learning (A. Krause & F. Yang): Led exercise sessions, developed and reviewed homework, and provided academic support in foundational machine learning topics.

Publications

Unraveling the formation of Tb/Pd films by coupled reduction as targets for heavy ion-beam irradiation

- <https://doi.org/10.1016/j.jallcom.2024.176954>

Projects

Brain Age Prediction

- Developed a regression model to estimate brain age from MRI-derived features

Heart Rhythm Classification from ECG Data

- Built a CNN based classifier for identifying heart rhythm anomalies from ECG signals

Others

- Sentiment Analysis with Fine-Tuned BERT
- Contrastive Learning for Image Analysis
- Image Classification with SWAG (Stochastic Weight Averaging-Gaussian)